

2.3 Characterizations of Invertible Matrices

Invertible Linear Transformations

A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be invertible if there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$S(T(\vec{x})) = \vec{x} \quad \text{for all } \vec{x} \text{ in } \mathbb{R}^n \quad (1)$$

$$T(S(\vec{x})) = \vec{x} \quad \text{for all } \vec{x} \text{ in } \mathbb{R}^n \quad (2)$$

The next theorem shows that if such an S exists, it is unique and must be a linear transformation. We call S the inverse of T and write it as T^{-1} .

Theorem 9

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(\vec{x}) = A^{-1} \vec{x}$ is the unique function satisfying equations (1) and (2).